SPECIFICATION — NODE 5 — PAGE 4

UPPER-BODY CONTEXT NODE — NODE-SPECIFIC EXEMPLAR CLAIM-LIKE STATEMENTS (OPTIONAL, NON-LIMITING)

A. Overview

This page provides optional, non-limiting claim-like statements for Node 5 to support standalone protectability of the upper-body context node. A provisional application does not require claims. These statements are illustrative only and do not limit the disclosure.

B. Exemplary Node 5 System Statements (Non-Limiting)

1. An upper-body wearable system comprising: a) a wearable device configured as a chest device, ring, or wristband; b) at least one cardiac sensing modality comprising ECG electrodes and/or optical pulse sensing (PPG); c) an inertial measurement unit (IMU) configured to generate motion features; and d) a computing system configured to derive context tags and confidence indicators; wherein the computing system uses the context tags to gate validity windows and condition interpretation of physiologic metrics. 2. The system of statement 1, wherein the computing system derives HRV features and suppresses HRV reporting during high-motion windows identified by IMU-based gating. 3. The system of statement 1, further comprising EDA/skin conductance sensing configured to derive autonomic activation features. 4. The system of statement 1, further comprising at least one temperature sensor configured to provide thermal context and coupling plausibility checks. 5. The system of statement 1, wherein the computing system computes a wear integrity index using at least one of electrode impedance checks, optical coupling checks, or capacitive proximity sensing. 6. The system of statement 1, wherein the computing system derives respiratory dynamics using chest motion patterns, inertial patterns, impedance-based respiration, or equivalent methods. 7. The system of statement 1, wherein the computing system outputs privacy-preserving context summaries comprising indices, tags, and confidence indicators without explicit anatomical imagery and without explicit anatomical labeling.

C. Exemplary Node 5 Method Statements (Non-Limiting)

8. A method comprising: a) acquiring upper-body sensor signals including at least one of ECG, PPG, IMU, temperature, and EDA; b) segmenting the sensor signals into analysis windows; c) computing at least one quality index including motion stability and wear integrity; d) deriving context tags including activity state and autonomic activation state; and e) outputting privacy-preserving context summaries with confidence indicators. 9. The method of statement 8, further comprising selecting matched windows for cross-device correlation only when the wear integrity index and motion stability index exceed thresholds. 10. The method of statement 8, further comprising computing a confidence-weighted conditioning factor used to adjust correlation scores among multiple nodes under consent gating.

D. Optional Integration Statements (Non-Limiting)

11. The system or method of any prior statement, further comprising time-aligning Node 5 context tags with a second device’s physiologic windows and excluding windows where Node 5 indicates high motion or poor wear integrity. 12. The system or method of any prior statement, further comprising applying context-specific baselines (rest-state vs. active-state) to normalize derived metrics generated by another node.

E. Variations

Any statement herein may be modified to include additional sensors, fewer sensors, substituted equivalent modalities, different form factors, different windowing methods, and different processing distributions (on-node, on-device, cloud) while maintaining the functional intent of capturing upper-body context and producing confidence-weighted gating/conditioning outputs.